

API Code Integration for NAMASTE Portal & ICD-11 TM2



A

Project Report

Submitted in partial fulfillment of the requirement for the award of degree of

Bachelor of Technology

In

Information Technology

Submitted to

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Declaration

We hereby declared that the work, which is being presented in the project entitled **API Code Integration for NAMASTE Portal & ICD-11 TM2** partial fulfillment of the requirement for the award of the degree of **Bachelor of Technology**, submitted in the department of Information Technology at **Acropolis Institute of Technology And Research, Indor** is an authentic record of my own work carried under the supervision of “**Prof. (Dr) Manish Vyas**”. I have not submitted the matter embodied in this report for the award of any other degree.

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Abstract

The healthcare ecosystem in India consists of both traditional AYUSH systems such as Ayurveda, Siddha, and Unani, as well as modern biomedical healthcare practices. However, these systems use different diagnostic terminology standards, creating challenges in interoperability, healthcare data exchange, Electronic Medical Record (EMR) integration, and clinical documentation. The NAMASTE (National AYUSH Morbidity and Standardized Terminologies Electronic) framework is widely used for AYUSH diagnosis coding, whereas modern healthcare systems primarily rely on the World Health Organization's ICD-11 classification standards. The absence of an effective integration mechanism between these terminologies limits seamless communication and standardized healthcare management.

This project proposes a FHIR-compliant API-based integration framework that enables interoperability between NAMASTE terminology and ICD-11 Traditional Medicine Module 2 (TM2). The proposed system supports dual-coded healthcare documentation, allowing both traditional and biomedical diagnoses to be recorded within a unified Electronic Medical Record. The framework utilizes standardized healthcare resources, terminology mapping services, RESTful APIs, and secure authentication mechanisms to facilitate efficient exchange of healthcare information.

The system is designed using a scalable microservice architecture and incorporates HL7 FHIR R4 standards to ensure compatibility with modern healthcare applications. It provides terminology lookup, code mapping, FHIR bundle generation, and secure access control for healthcare data management. By automating the mapping process between traditional and biomedical terminologies,

The proposed solution contributes to the modernization of AYUSH healthcare infrastructure and supports national digital health initiatives by enabling standardized, secure, and interoperable healthcare records. Furthermore, it creates opportunities for improved healthcare analytics, clinical research, insurance claim processing, and future integration of intelligent technologies such as artificial intelligence and advanced decision-support systems.

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Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
CNN	Convolutional Neural Network
DFD	Data Flow Diagram
DL	Deep Learning
ER	Entity Relationship
FA	Fundamental Analysis
FinBERT	Financial Bidirectional Encoder Representations from Transformers
LSTM	Long Short-Term Memory
MA	Moving Average
MACD	Moving Average Convergence Divergence
ML	Machine Learning
NLP	Natural Language Processing
OHLCV	Open, High, Low, Close, Volume
RSI	Relative Strength Index
TA	Technical Analysis

UI	User Interface
XGBoost	Extreme Gradient Boosting

Chapter 1

Introduction

1.1 Rationale

Healthcare Digitalization

The healthcare sector is rapidly adopting digital technologies to improve patient care, medical record management, and healthcare data exchange. Electronic Medical Records (EMRs) and healthcare information systems have become essential components of modern healthcare infrastructure, enabling healthcare providers to manage patient information more efficiently.

Need for Interoperability

India has a unique healthcare ecosystem where traditional AYUSH systems such as Ayurveda, Siddha, and Unani coexist with modern biomedical healthcare practices. However, these systems use different diagnostic terminology standards. AYUSH practitioners use the NAMASTE (National AYUSH Morbidity and Standardized Terminologies Electronic) framework, while modern healthcare institutions primarily rely on the World Health Organization's ICD-11 classification system. This difference creates challenges in maintaining unified patient records and exchanging healthcare information across different healthcare platforms.

Motivation for the Proposed System

Healthcare interoperability standards such as HL7 FHIR provide an effective approach for integrating different healthcare systems. Therefore, there is a need for a standardized framework that can connect NAMASTE terminology with ICD-11 codes and support seamless healthcare data exchange. The proposed system aims to improve clinical documentation, support EMR integration, and promote secure and standardized healthcare interoperability between traditional and modern healthcare systems.

1.2 Existing System

Current Healthcare Documentation Systems

At present, most healthcare organizations use Electronic Medical Record (EMR) systems to store and manage patient information digitally. These systems are primarily designed

around modern biomedical standards such as ICD-10 and ICD-11, enabling standardized disease classification, clinical documentation, and healthcare reporting. However, support for traditional AYUSH terminologies remains limited in many existing healthcare platforms.

Existing AYUSH and ICD-11 Platforms

The NAMASTE portal provides standardized terminology and morbidity codes for AYUSH systems, helping practitioners maintain consistent clinical records. Similarly, the WHO ICD-11 platform offers internationally accepted disease classification standards, including the Traditional Medicine Module 2 (TM2). Although both platforms serve important purposes, they operate independently and do not provide a direct mechanism for seamless integration or dual-coded healthcare documentation.

Limitations of Existing Systems

Most existing EMR systems do not support automatic mapping between NAMASTE and ICD-11 codes. As a result, healthcare professionals often need to manage traditional and biomedical diagnoses separately, leading to duplicate efforts, fragmented records, and reduced interoperability. In addition, the absence of a unified terminology integration framework limits healthcare analytics, insurance claim processing, and effective exchange of healthcare information across different healthcare systems.

Need for Improvement

These limitations highlight the need for a standardized integration framework that can connect NAMASTE and ICD-11 terminologies through modern interoperability standards such as HL7 FHIR. Such a solution can support dual coding, improve healthcare data consistency, and enable seamless integration with existing EMR systems.

1.3 Problem Formulation

Existing Problem

The Indian healthcare ecosystem consists of both traditional AYUSH systems and modern biomedical healthcare practices. While both play an important role in patient care, they use different diagnostic coding standards. AYUSH practitioners use NAMASTE terminology, whereas modern healthcare systems rely on ICD-11

classifications. Due to this difference, healthcare information cannot be exchanged efficiently between the two systems.

Challenges Faced

Most existing Electronic Medical Record (EMR) systems are designed primarily for biomedical healthcare data and provide limited support for AYUSH terminology. This makes it difficult to record traditional and biomedical diagnoses together within a single patient record. As a result, healthcare providers face challenges in maintaining standardized documentation, generating healthcare reports, and ensuring interoperability between different healthcare platforms.

Need for a Solution

With the increasing adoption of digital healthcare systems and national initiatives such as the Ayushman Bharat Digital Mission (ABDM), there is a growing need for a framework that can integrate NAMASTE and ICD-11 terminologies. Such a system should support standardized data exchange, dual-coded diagnosis recording, secure access, and seamless EMR integration while complying with healthcare interoperability standards.

Problem Statement

The problem addressed in this project is the lack of interoperability between NAMASTE and ICD-11 terminology systems in existing healthcare applications. The objective is to develop a FHIR-compliant API-based integration framework that enables terminology mapping, dual-coded healthcare documentation, secure healthcare data exchange, and seamless integration with Electronic Medical Record systems. This framework aims to improve healthcare interoperability, clinical documentation, and data standardization across traditional and modern healthcare domains.

1.4 Proposed System

Overview of the Proposed System

The proposed system is a FHIR-compliant API-based integration framework designed to establish interoperability between NAMASTE and ICD-11 TM2 terminology systems. The framework enables healthcare organizations to record, manage, and exchange both traditional and biomedical diagnoses in a standardized digital format. By providing a common platform for terminology integration, the system improves healthcare

documentation and supports seamless communication between different healthcare applications.

Key Components

The system consists of several functional components including a terminology lookup service, terminology mapping module, FHIR bundle generation service, authentication mechanism, and healthcare data management layer. These components work together to retrieve healthcare terminologies, perform code mapping, generate standardized healthcare records, and facilitate secure data exchange.

Working of the System

Healthcare practitioners can search for a NAMASTE diagnosis term through the system interface. The framework then identifies the corresponding ICD-11 TM2 concept and provides dual-coded diagnostic information. The generated data is converted into standardized HL7 FHIR resources, allowing seamless integration with Electronic Medical Record (EMR) systems. Secure access is maintained using authentication and authorization mechanisms to ensure data privacy and regulatory compliance.

Expected Benefits

The proposed system aims to reduce manual coding efforts, improve documentation accuracy, and enhance healthcare interoperability. It supports standardized healthcare data exchange, better clinical record management, healthcare analytics, insurance claim processing, and future digital health initiatives. By connecting traditional and modern healthcare terminology systems, the framework contributes to a more integrated healthcare ecosystem.

diagram

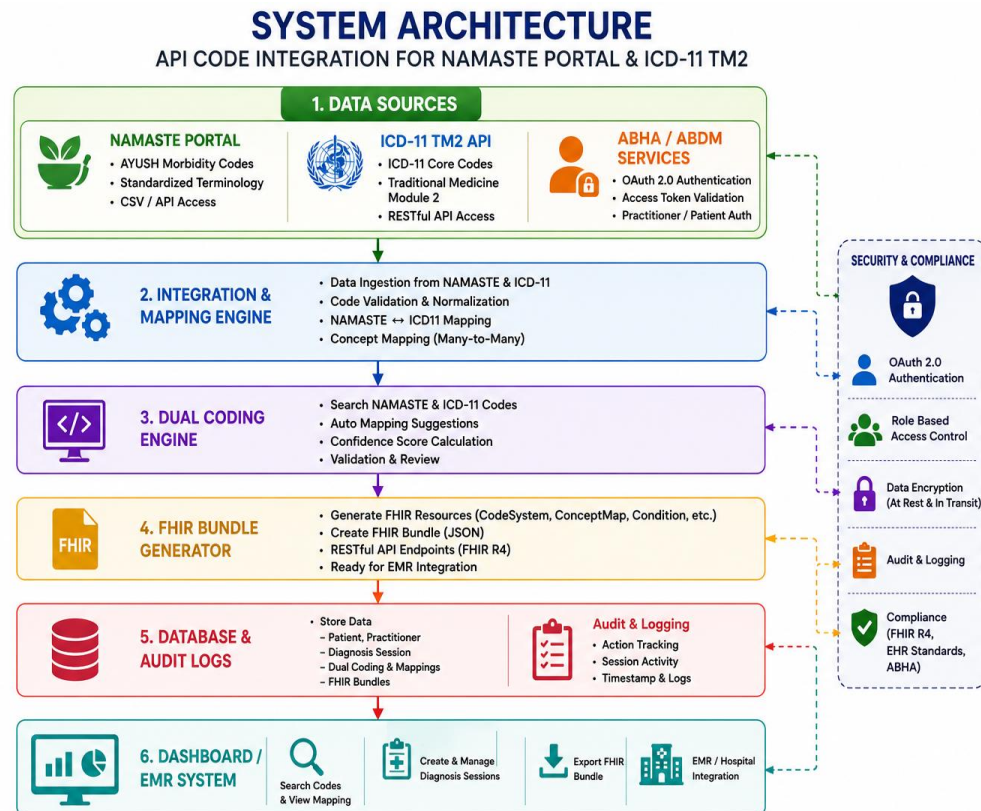


Fig 1 full system process

1.5 Objectives

Main Objective

The main objective of this project is to develop a FHIR-compliant API-based integration framework that enables interoperability between NAMASTE and ICD-11 TM2 terminology systems. The system aims to support standardized healthcare documentation, dual-coded diagnosis recording, and seamless integration with Electronic Medical Record (EMR) systems while complying with modern healthcare standards.

Specific Objectives

- To integrate NAMASTE terminology with WHO ICD-11 TM2 and biomedical disease classifications.

- To support dual-coded healthcare documentation for both traditional and modern medical diagnoses.
- To provide standardized terminology mapping and code translation services.
- To generate HL7 FHIR R4 compliant healthcare records for EMR integration.
- To implement secure authentication and controlled access for healthcare data exchange.
- To improve healthcare interoperability, data consistency, and clinical documentation accuracy.
- To support healthcare analytics, research activities, and insurance claim processing through standardized coding systems.

Expected Outcome

The successful implementation of the proposed system will facilitate efficient healthcare data exchange between traditional AYUSH and biomedical healthcare systems. It will improve interoperability, reduce manual documentation efforts, and contribute towards the development of a unified digital healthcare ecosystem.

1.6 Contribution of the Project

- **Technical Contribution**
 - This project contributes to healthcare interoperability by providing a standardized framework for integrating NAMASTE and ICD-11 TM2 terminology systems. The proposed solution supports terminology mapping, dual-coded diagnosis recording, and HL7 FHIR-based healthcare data exchange, enabling seamless communication between different healthcare applications.
- **Healthcare Contribution**
 - The system helps improve clinical documentation by allowing traditional AYUSH and biomedical diagnoses to be recorded in a unified format. This reduces data inconsistency, enhances record management, and supports better healthcare reporting and analytics.
- **Research and Future Contribution**
 - The framework creates a foundation for future advancements in digital healthcare, including intelligent terminology mapping, healthcare analytics, decision-support

systems, and broader integration with national healthcare initiatives. It also promotes the adoption of standardized healthcare practices across different medical domains.

-

1.6.1 Market Potential

Growing Digital Healthcare Market

The healthcare industry is rapidly adopting digital technologies to improve patient care, record management, and healthcare services. With the increasing use of Electronic Medical Records (EMRs) and digital health platforms, the demand for interoperable healthcare solutions is continuously growing.

Expansion of AYUSH Healthcare

India has a large AYUSH healthcare network consisting of hospitals, clinics, and practitioners. As the sector moves toward digital transformation, there is a growing need for standardized systems that can integrate traditional medicine records with modern healthcare infrastructure. Such integration can improve healthcare accessibility, documentation quality, and patient management.

Opportunities for Healthcare Integration

The proposed framework can be utilized by hospitals, EMR vendors, healthcare organizations, insurance providers, and government healthcare programs. By enabling standardized terminology mapping and healthcare data exchange, it supports efficient clinical workflows and better healthcare reporting.

Long-Term Potential

With the increasing focus on healthcare interoperability and national digital health initiatives, integrated healthcare solutions are expected to play an important role in future healthcare systems. The proposed framework provides a foundation for scalable healthcare applications, advanced analytics, and intelligent healthcare services.

1.6.2 Innovativeness

Integration of Traditional and Modern Healthcare

The project introduces an innovative approach by connecting NAMASTE terminology used in AYUSH systems with WHO ICD-11 TM2 standards. This integration helps bridge the gap between traditional and biomedical healthcare systems within a single digital framework.

Dual-Coding Mechanism

Unlike conventional healthcare applications that support only a single coding standard, the proposed system enables dual-coded diagnosis recording. This allows healthcare professionals to document both traditional and biomedical perspectives of a disease within the same patient record.

FHIR-Based Interoperability

The use of HL7 FHIR standards provides a modern and standardized approach for healthcare data exchange. The framework supports seamless communication between different healthcare systems while maintaining compatibility with existing EMR platforms.

Secure and Scalable Architecture

The proposed framework combines interoperability, terminology mapping, secure authentication, and standardized healthcare records within a scalable architecture. This makes the system adaptable to future healthcare requirements and digital health initiatives.

1.6.3 Usefulness

Improved Clinical Documentation

The proposed system enables healthcare professionals to record both traditional AYUSH and biomedical diagnoses in a standardized format. This improves the accuracy and consistency of patient records across different healthcare systems.

Enhanced Healthcare Interoperability

By integrating NAMASTE and ICD-11 TM2 terminologies, the framework facilitates seamless healthcare data exchange between hospitals, clinics, and Electronic Medical Record (EMR) systems. This reduces communication barriers and improves coordination among healthcare providers.

Support for Healthcare Analytics

Standardized and structured healthcare data can be used for disease analysis, healthcare research, and policy planning. The system helps generate reliable healthcare information that can support decision-making at organizational and national levels.

Benefits for Patients and Healthcare Organizations

The framework simplifies record management, reduces manual coding efforts, and improves overall healthcare efficiency. It also supports insurance claim processing and promotes better utilization of digital healthcare services.

1.7 Report Organization

Table No.1 Report Organization

The report is organized into different chapters to provide a systematic understanding of the project. Each chapter focuses on a specific aspect of the proposed system, beginning with the project background and ending with the final conclusions and future enhancements.

Chapter	Title	Description
Chapter 1	Introduction	Presents the project background, rationale, problem formulation, proposed system, objectives, and contribution of the project.
Chapter 2	Requirement Engineering	Discusses feasibility analysis, requirement collection, functional and non-functional requirements, hardware and software requirements, and use-case diagrams.
Chapter 3	Analysis & Conceptual Design	Describes the system architecture, UML diagrams, data flow diagrams, database design, and user interface design.
Chapter 4	Implementation & Testing	Explains the implementation methodology, technologies used, system development process, and testing approaches.

Chapter 5	Results & Discussion	Presents system outputs, interface snapshots, database details, observations, and final findings obtained from the project.
Chapter 6	Conclusion & Future Scope	Summarizes the overall work, achievements of the project, and possible future enhancements.

The report follows a structured approach to explain the design, development, implementation, and evaluation of the proposed healthcare interoperability framework.

Chapter 2

Requirement Engineering

2.1 Feasibility Study

The feasibility study evaluates whether the proposed healthcare interoperability framework can be successfully developed and implemented using available technologies and resources. The project focuses on integrating NAMASTE and ICD-11 TM2 terminologies through standardized healthcare technologies to improve healthcare data exchange and EMR integration. Based on the analysis, the proposed system is considered feasible and suitable for practical implementation

2.1.1 Technical Feasibility

The proposed system is technically feasible due to the availability of modern healthcare standards and software technologies. The following factors support its implementation:

- HL7 FHIR R4 provides a standardized framework for healthcare data exchange and interoperability.
- RESTful APIs can be used for communication between NAMASTE, ICD-11 TM2, and EMR systems.
- Database technologies such as PostgreSQL and MongoDB can efficiently store healthcare records, terminology data, and audit logs.

- WHO ICD-11 resources and NAMASTE terminology datasets are available for terminology mapping and integration.
- OAuth 2.0 and ABHA-based authentication mechanisms can be used to ensure secure access to healthcare information.
- Existing EMR systems support API-based integration, making deployment and adoption easier.
- Modern development technologies such as React.js, Node.js, and cloud platforms provide a scalable and maintainable development environment.

Therefore, the required technologies, standards, and resources are available, making the development and implementation of the proposed framework technically achievable

2.1.2 Economical Feasibility

The proposed system is economically feasible because it can be developed using widely available open-source technologies and healthcare standards. The following factors support its economic viability:

- Open-source technologies such as React.js, Node.js, PostgreSQL, and MongoDB reduce software development costs.
- HL7 FHIR standards and RESTful APIs can be implemented without expensive proprietary solutions.
- Existing healthcare infrastructure and EMR systems can be integrated with minimal modifications.
- Cloud-based deployment options help reduce hardware and maintenance expenses.
- The system minimizes manual documentation efforts, reducing operational costs for healthcare organizations.
- Standardized healthcare records improve efficiency in healthcare management, analytics, and insurance claim processing.
- The framework can be scaled according to organizational requirements without significant additional investment.

2.1.3 Operational Feasibility

Ease of Implementation

The proposed system can be integrated with existing healthcare applications and Electronic Medical Record (EMR) systems without major modifications. Its standardized architecture makes deployment and adoption easier for healthcare organizations.

User Acceptance

- Provides a user-friendly interface for terminology search and diagnosis mapping.
- Supports healthcare professionals in recording both traditional and biomedical diagnoses.
- Requires minimal training for effective utilization.
- Reduces manual efforts involved in healthcare documentation and code management.

Operational Benefits

- Improves accuracy and consistency of healthcare records.
- Enables secure and standardized healthcare data exchange.
- Supports interoperability between different healthcare systems.
- Assists healthcare organizations in complying with digital health standards and initiatives.
-

2.2 REQUIREMENT COLLECTION

2.2.1 Discussion

Understanding the Problem

A detailed study was conducted to understand the challenges associated with healthcare interoperability between traditional AYUSH systems and modern biomedical healthcare systems. It was observed that different terminology standards create difficulties in maintaining unified healthcare records and exchanging healthcare information.

Analysis of Existing Systems

Existing healthcare applications, EMR systems, and terminology platforms were analyzed to identify their strengths and limitations. The analysis revealed that most systems support either NAMASTE or ICD-11 terminology but do not provide an efficient mechanism for integrating both standards within a single healthcare framework.

Stakeholder Requirements

The requirements of healthcare professionals, healthcare organizations, and digital health platforms were considered during the discussion process. The need for standardized terminology mapping, secure healthcare data exchange, and dual-coded diagnosis recording was identified as a key requirement.

Discussion Outcome

The discussion process highlighted the necessity of a FHIR-based integration framework capable of connecting NAMASTE and ICD-11 TM2 terminologies. The findings served as the foundation for defining the system requirements and designing the proposed solution.

2.2.2 Requirement Analysis

Requirement Identification

- The collected requirements were analyzed to determine the functionalities and features necessary for the proposed system. The analysis focused on healthcare interoperability, terminology integration, secure data exchange, and compliance with healthcare standards.

Functional Requirements Analysis

- The system should be capable of integrating NAMASTE and ICD-11 TM2 terminologies, performing terminology mapping, generating FHIR-compliant healthcare records, and supporting secure communication between healthcare applications. It should also provide terminology search and dual-coded diagnosis recording facilities.

Non-Functional Requirements Analysis

- The framework should ensure security, reliability, scalability, and interoperability. It must support secure authentication, efficient performance, easy maintenance, and compatibility with existing Electronic Medical Record (EMR) systems.

Analysis Outcome

- The requirement analysis helped define the core functionalities and quality attributes of the proposed framework. These requirements serve as the foundation for system design, implementation, and testing.

2.3 REQUIREMENTS

2.3.1 Functional Requirements

Table No.2 Functional Requirements

Functional requirements describe the specific functions and services that the proposed system must perform. These requirements define the core operations necessary for integrating NAMASTE and ICD-11 TM2 terminologies and supporting healthcare interoperability.

Functional Requirements

FR No.	Functional Requirement	Description
FR1	Terminology Integration	Integrate NAMASTE and ICD-11 TM2 terminology systems.
FR2	Terminology Search	Allow users to search healthcare terms and codes.
FR3	Code Mapping	Map NAMASTE terms to corresponding ICD-11 TM2 concepts.
FR4	Dual Coding Support	Enable recording of both traditional and biomedical diagnoses.
FR5	FHIR Bundle Generation	Generate HL7 FHIR-compliant healthcare records.
FR6	Authentication	Provide secure user authentication and authorization.
FR7	EMR Integration	Support integration with Electronic Medical Record systems.

FR8	Audit Logging	Maintain logs of system activities and transactions.
FR9	Data Exchange	Enable secure healthcare data sharing through APIs.

Purpose

These functional requirements ensure that the proposed framework can effectively support terminology integration, healthcare documentation, and interoperability between different healthcare systems.

NFR No.	Non-Functional Requirement	Description
NFR1	Performance	The system should provide fast terminology search and code mapping services.
NFR2	Security	Healthcare data should be protected through secure authentication and access control mechanisms.
NFR3	Reliability	The framework should provide accurate and consistent healthcare data processing.
NFR4	Scalability	The system should support future expansion and increased healthcare data volume.
NFR5	Availability	Services should remain accessible whenever required by authorized users.
NFR6	Maintainability	The system should be easy to update, manage, and maintain.
NFR7	Interoperability	The framework should support standardized communication with different healthcare

		applications.
NFR8	Usability	The interface should be simple and easy to use for healthcare professionals.

2.3.1.1 Statement of Functionality

2.3.1.1 Statement of Functionality

The proposed system provides a standardized framework for integrating NAMASTE and ICD-11 TM2 terminology systems within healthcare applications. It enables healthcare professionals to search medical terminologies, perform code mapping, record dual-coded diagnoses, and generate FHIR-compliant healthcare records.

The framework supports secure authentication, healthcare data exchange, and integration with Electronic Medical Record (EMR) systems through standardized APIs. Additionally, it maintains audit logs and facilitates efficient healthcare documentation while ensuring interoperability between traditional AYUSH and modern biomedical healthcare systems.

2.3.2 Non-Functional Requirements

- Non-functional requirements define the quality attributes and operational characteristics of the proposed system. These requirements ensure that the framework performs efficiently, remains secure, and provides a reliable user experience.
- **Non-Functional Requirements**

2.3.2.1 Statement of Functionality (Non-Functional Summary)

The proposed system is designed to provide secure, reliable, and efficient healthcare interoperability services. It ensures fast terminology search, accurate code mapping, secure healthcare data exchange, and seamless integration with Electronic Medical Record (EMR) systems.

The framework supports scalability, maintainability, and interoperability through the use of standardized healthcare technologies and APIs. It also provides a user-friendly environment for healthcare professionals while ensuring the confidentiality, integrity, and availability of healthcare information.

2.4 Hardware & Software Requirements

This section outlines the system requirements for development and for end-users who will interact with the prediction dashboard.

2.4.1 Hardware Requirement

The proposed system requires adequate hardware resources for development, testing, deployment, and daily operation. These resources ensure smooth execution of healthcare data processing, terminology mapping, and API services.

Developer Hardware Requirements

Component	Requirement
Processor	Intel Core i5 / AMD Ryzen 5 or higher
RAM	Minimum 8 GB
Storage	256 GB SSD or higher
Internet Connection	Stable Broadband Connection
Display	Standard HD Monitor

End-User Hardware Requirements

Component	Requirement
------------------	--------------------

Processor	Dual Core Processor or higher
RAM	Minimum 4 GB
Storage	20 GB Free Space
Internet Connection	Stable Internet Connection
Device	Desktop, Laptop, or Tablet

2.4.2 Software Requirement

Developer Software Requirements

Table No.5 Developer Software RequirementsThe proposed healthcare interoperability framework requires various software tools and technologies for development, testing, deployment, and maintenance. These software resources support healthcare data management, API integration, security, and user interface development.

Developer Software Requirements

Software Category	Requirement
Operating System	Windows 10/11, Linux, or macOS
Programming Languages	JavaScript, Python
Frontend Technologies	HTML, CSS, React.js
Backend Technologies	Node.js, Express.js

Database	PostgreSQL, MongoDB
API Testing Tool	Postman
Version Control	Git, GitHub
Development Environment	Visual Studio Code

End-User Software Requirements

Software Category	Requirement
Operating System	Windows, Linux, macOS
Web Browser	Google Chrome, Mozilla Firefox, Microsoft Edge
Internet Connection	Required for accessing healthcare services
EMR System	Any FHIR-compatible healthcare application (if integrated)

2.5 Use-case Diagrams

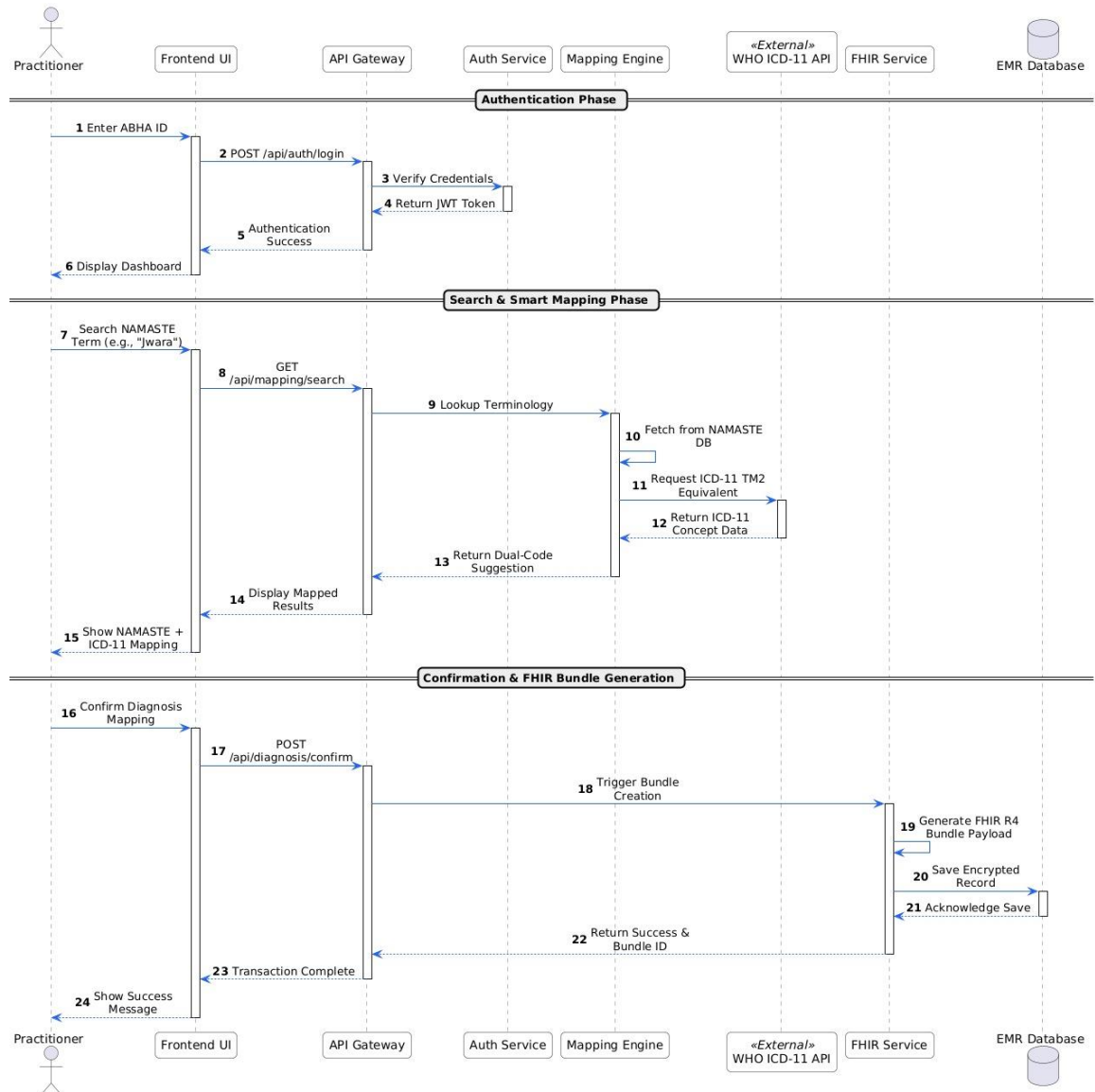


Fig 2 Use-Case Diagrams

Figure 2: Use-case diagram shows primary actor *User* and external data actors (External APIs). Core system use cases (center) are:

- **Authenticate / Manage Profile** (login, register, update settings)
- **Search Stock → Analyze Stock** (the central use case)
- **Analyze Stock** includes: view price predictions, view technical analysis, view fundamental analysis, view market sentiment.

- **Get Buy/Sell/Hold Recommendation** (includes explanation and confidence), **Set Price Alerts, View Explanation (SHAP/feature drivers)**.

External APIs supply price / news / fundamentals.

2.5.1 Use-case Descriptions

User Authentication

This use case allows healthcare practitioners and administrators to securely access the system using authorized credentials. The authentication mechanism verifies user identity before granting access to healthcare services and resources.

Search NAMASTE Terminology

Healthcare practitioners can search for NAMASTE terms and codes through the system interface. The system retrieves relevant terminology information from the database and displays matching results.

Retrieve ICD-11 TM2 Codes

The framework communicates with ICD-11 resources to obtain corresponding traditional medicine and biomedical classification codes. This helps users access standardized healthcare terminologies.

Perform Terminology Mapping

The system maps NAMASTE terms with their corresponding ICD-11 TM2 concepts. This functionality supports dual-coded healthcare documentation and improves interoperability.

Generate FHIR Bundle

After successful terminology mapping, the system generates standardized HL7 FHIR-compliant healthcare records. These records can be exchanged with other healthcare applications and systems.

Submit Data to EMR System

The generated healthcare records are transmitted to Electronic Medical Record (EMR) systems through secure APIs. This ensures proper storage and management of patient information.

Manage Terminology Database

The proposed system follows a FHIR-compliant microservice architecture designed to enable interoperability between NAMASTE and ICD-11 TM2 terminology systems. The architecture consists of multiple independent components that work together to provide secure healthcare data exchange, terminology mapping, and EMR integration.

Frontend Layer

The frontend layer provides an interface through which healthcare practitioners can search medical terminologies, perform diagnosis mapping, and generate healthcare records. It acts as the communication point between users and the system.

API Gateway Layer

The API Gateway manages incoming requests and routes them to the appropriate backend services. It also handles request validation, security checks, and communication between different system components.

Authentication Service

This component is responsible for user authentication and authorization. It verifies user credentials and ensures that only authorized users can access healthcare resources and services.

Terminology Mapping Service

The terminology mapping service performs the core functionality of the system by mapping NAMASTE terms to corresponding ICD-11 TM2 concepts. It helps support dual-coded healthcare documentation and interoperability.

FHIR Bundle Generation Service

This service generates standardized HL7 FHIR-compliant healthcare records containing patient information, diagnoses, and healthcare encounter details for seamless EMR integration.

Database Layer

The database layer stores healthcare terminologies, mapping information, healthcare records, user data, and audit logs. It ensures secure and efficient management of healthcare information.

3.2 Sequence Diagrams

The Sequence Diagram illustrates the interaction between different components of the proposed system during a healthcare documentation process. It shows how a healthcare practitioner authenticates through the system, searches for a NAMASTE term, and requests the corresponding ICD-11 TM2 mapping. The system then communicates with the terminology services, generates a FHIR-compliant healthcare record, and submits the information to the EMR system. This diagram helps visualize the order of messages exchanged between system components and external healthcare services.

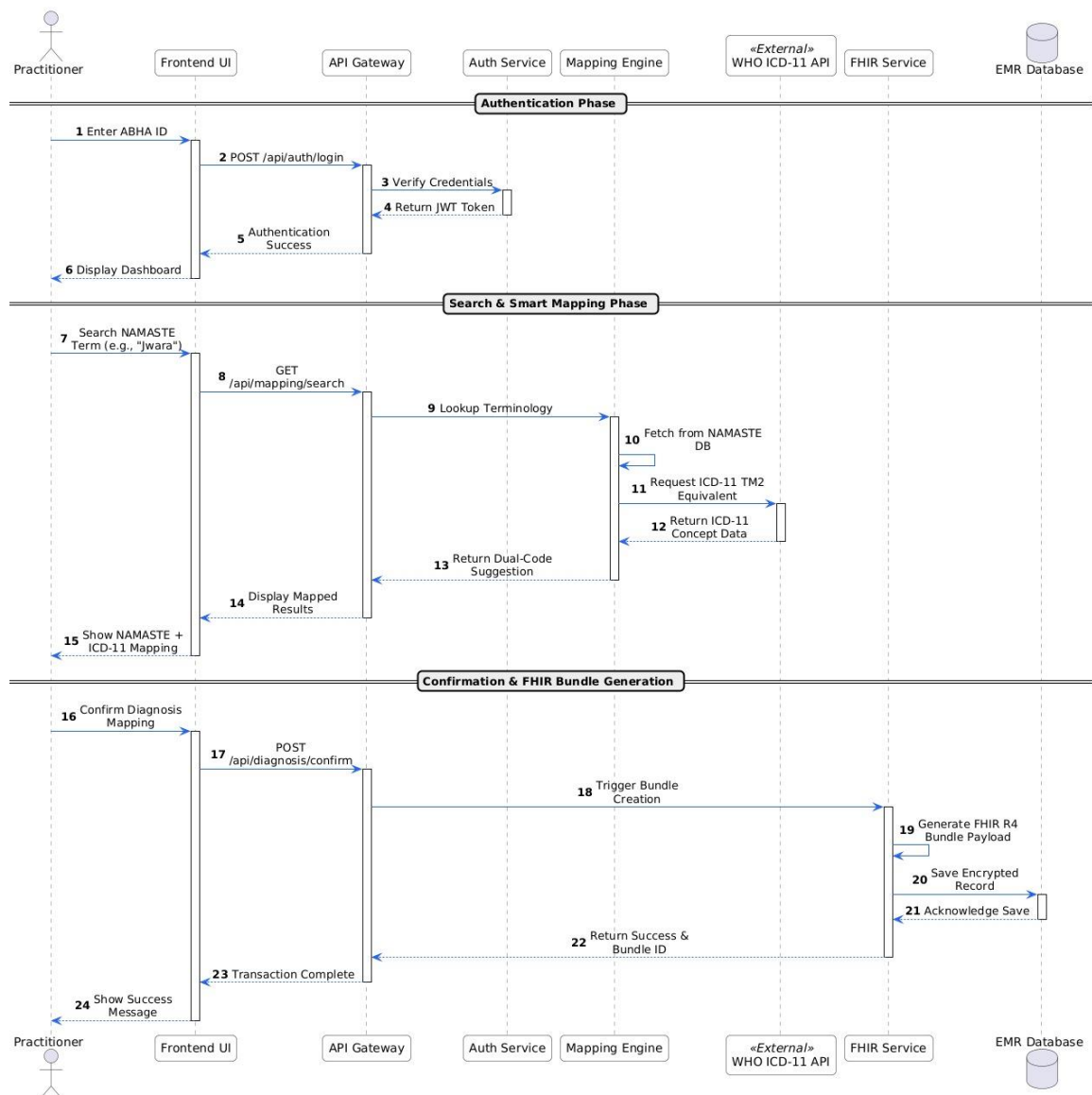


Fig 4 Sequence Diagrams

3.3 Class Diagrams

The Class Diagram represents the static structure of the proposed system by showing the main classes, their attributes, methods, and relationships. It includes classes such as Practitioner, Authentication Service, Terminology Mapping Service, ICD-11 Concept, FHIR Bundle Generator, and Audit Logger. The diagram helps define how system components are organized and interact with each other. It provides a clear blueprint for system development and implementation.

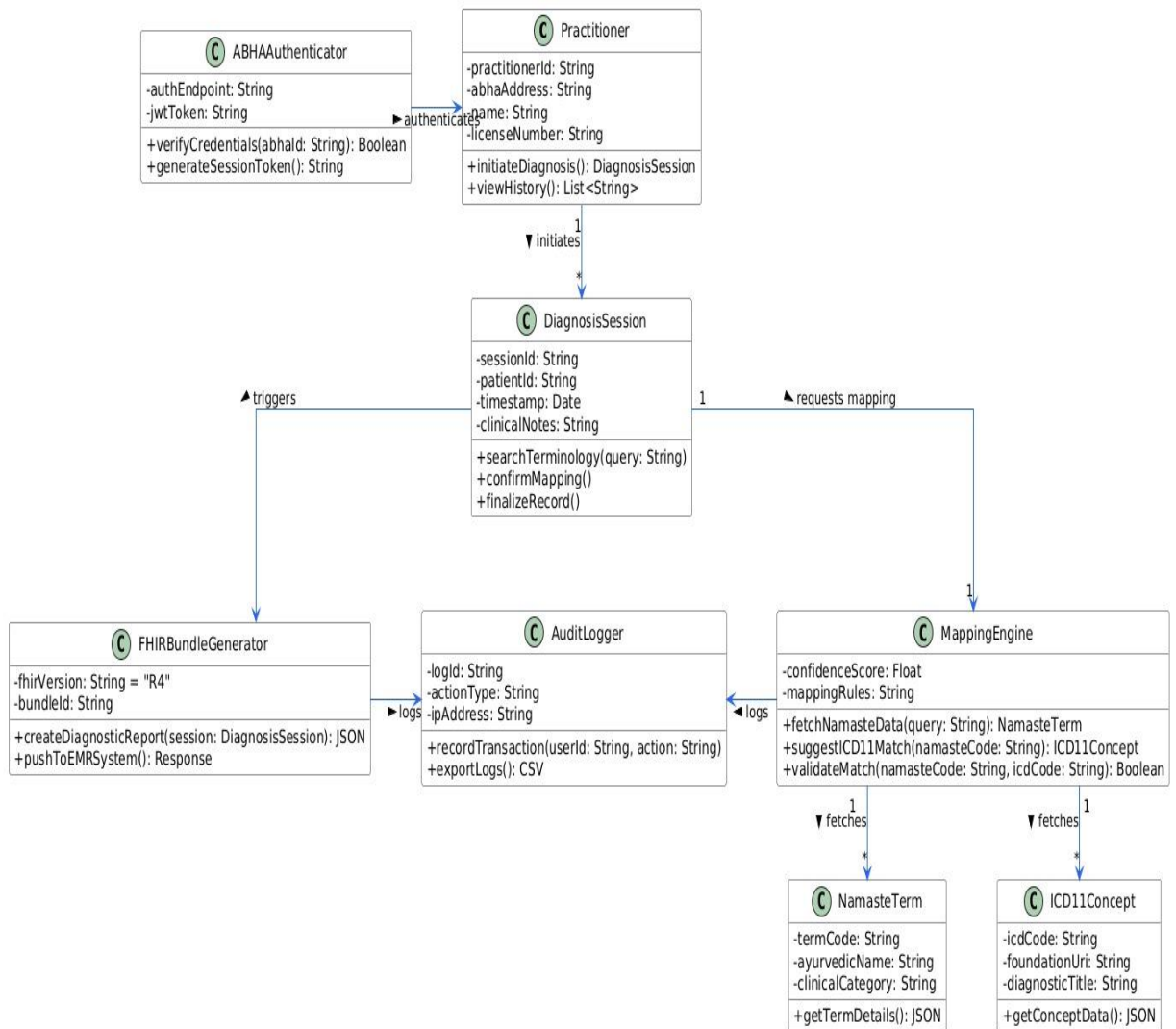


Fig 5 Class Diagram

3.4 Data Flow Diagrams

3.4 Data Flow Diagram (DFD)

The Data Flow Diagram (DFD) illustrates how healthcare data moves through different components of the proposed system. It shows the flow of information from healthcare practitioners to the terminology mapping services, databases, and EMR systems. The DFD helps identify data processing activities, data storage locations, and external entities interacting with the framework. It provides a clear understanding of how healthcare information is processed and exchanged within the system.

3.5 User Interface Design

The User Interface (UI) is designed to provide a simple and user-friendly environment for healthcare practitioners. It allows users to search NAMASTE terms, view ICD-11 TM2 mappings, generate FHIR records, and manage healthcare documentation efficiently. The interface focuses on ease of use, quick navigation, and clear presentation of healthcare information. A well-designed UI improves user experience and enhances the overall efficiency of the system.

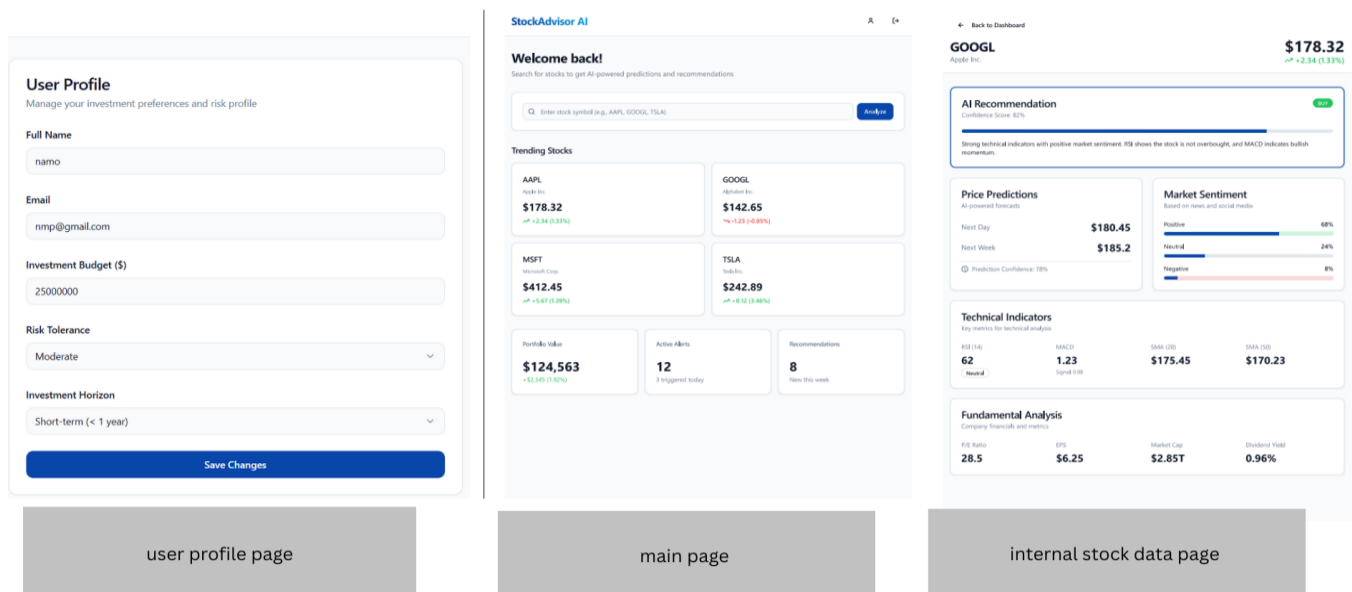


Fig 9 User Interface

3.6 Data Design

Data Design defines how healthcare information is stored, organized, and managed within the proposed system. It includes terminology data, user information, healthcare records, mapping details, and audit logs. A well-structured data design ensures efficient data retrieval, consistency, and security. It also supports smooth integration between NAMASTE, ICD-11 TM2, and Electronic Medical Record (EMR) systems.

3.6.1 Schema Definitions

The Schema Definition describes the structure of the database tables used in the proposed system. It defines entities such as users, healthcare records, terminology data, ICD-11 mappings, and audit logs along with their attributes and relationships. A proper schema ensures data consistency, efficient storage, and faster retrieval of healthcare information. It serves as the foundation for managing and exchanging healthcare data within the framework.

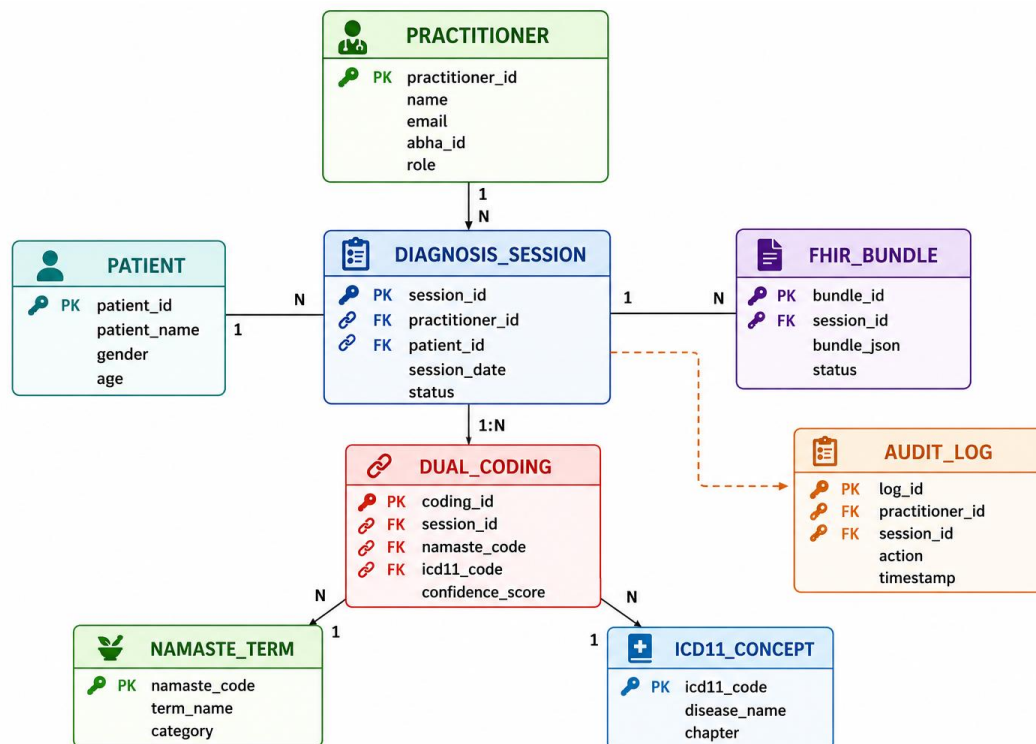


Fig 10 Schema Diagram

3.6.2 E-R Diagram

The Entity Relationship (ER) Diagram represents the relationships between the major entities of the proposed system. It illustrates how users, healthcare records, NAMASTE terminologies, ICD-11 concepts, FHIR records, and audit logs are connected within the database. The ER Diagram helps in understanding data dependencies and maintaining data integrity. It also provides a clear view of the database structure used in the healthcare interoperability framework.

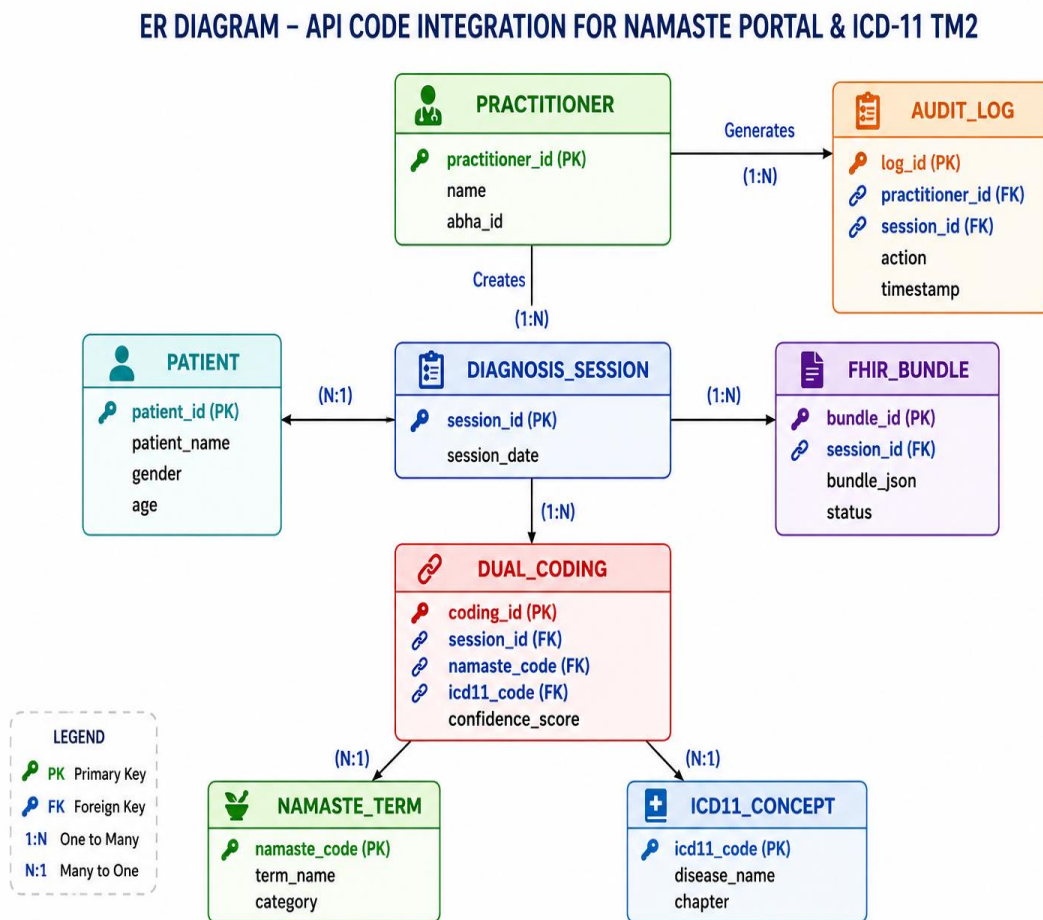


Fig 11 E-R Diagram

Chapter 4

Implementation & Testing

4.1 Methodology

The methodology of the proposed healthcare interoperability framework follows a structured approach to integrate NAMASTE and ICD-11 TM2 terminologies while ensuring secure healthcare data exchange and compliance with healthcare standards. The development process begins with the collection and analysis of healthcare terminology resources, interoperability standards, and system requirements. This helps in understanding the challenges associated with healthcare documentation and terminology integration.

The next phase focuses on designing a FHIR-compliant architecture capable of supporting terminology mapping, healthcare record generation, and secure communication between different healthcare systems. The architecture is divided into multiple components such as authentication services, terminology mapping services, FHIR bundle generation services, and database management systems. This modular approach improves scalability, maintainability, and system performance.

After the design phase, the implementation process is carried out using modern web technologies, databases, and healthcare interoperability standards. The system integrates NAMASTE terminology resources with ICD-11 TM2 concepts and provides standardized APIs for healthcare data exchange. Authentication and authorization mechanisms are also implemented to ensure secure access to healthcare information.

Once the development process is completed, testing and validation activities are performed to verify the functionality, security, and interoperability of the framework. Different system components are tested individually as well as collectively to ensure accurate terminology mapping, efficient healthcare record generation, and successful integration with Electronic Medical Record (EMR) systems.

The overall methodology provides a systematic approach for developing a secure, scalable, and interoperable healthcare integration framework capable of supporting standardized healthcare documentation and seamless healthcare data exchange.

4.1.1 Proposed Algorithm

The proposed system follows the following steps:

1. User Authentication

- The healthcare practitioner logs into the system using authorized credentials.
- The authentication service verifies the user's identity before granting access.

2. Terminology Search

- The user searches for a NAMASTE term through the system interface.
- The search request is processed by the terminology service.

3. Terminology Retrieval

- The system retrieves the corresponding NAMASTE terminology data from the database.
- Relevant healthcare information is displayed to the user.

4. Code Mapping

- The terminology mapping module identifies the corresponding ICD-11 TM2 concept.
- The mapped results are presented for verification.

5. Dual-Coded Diagnosis Creation

- The selected NAMASTE and ICD-11 TM2 codes are combined.
- A dual-coded healthcare diagnosis record is generated.

6. FHIR Record Generation

- The system creates a standardized HL7 FHIR-compliant healthcare record.
- Patient information and diagnosis details are included in the record.

7. Data Validation

- The generated healthcare record is validated according to interoperability standards.
- Any inconsistencies are checked before further processing.

8. EMR Integration

- The validated FHIR record is securely transmitted to the Electronic Medical Record (EMR) system.
- The healthcare information is stored for future reference.

9. Audit Logging

- All important system activities and transactions are recorded in the audit log.
- These logs support monitoring, security, and compliance requirements.

10. Process Completion

- The healthcare record is successfully stored and made available for authorized healthcare services.
-

4.2 Implementation Approach

The implementation phase focuses on developing the proposed healthcare interoperability framework using modern technologies and healthcare standards. The system is implemented using a modular architecture to ensure scalability, maintainability, and efficient integration with healthcare applications.

Frontend Implementation

- Developed using HTML, CSS, JavaScript, and React.js.
- Provides interfaces for terminology search, code mapping, and healthcare record management.
- Ensures easy navigation and user-friendly interaction.

Backend Implementation

- Developed using Node.js and Express.js.
- Handles business logic, API requests, authentication, and data processing.
- Facilitates communication between different system components.

Database Implementation

- PostgreSQL and MongoDB are used for storing healthcare records, terminology data, mappings, and audit logs.
- Ensures efficient storage and retrieval of healthcare information.

API & Integration Implementation

- RESTful APIs are used for communication between modules.
- HL7 FHIR standards are implemented for healthcare data exchange.
- Supports integration with Electronic Medical Record (EMR) systems.

Security Implementation

- OAuth 2.0 authentication mechanisms are used for secure access control.
- Audit logging is implemented to monitor system activities and maintain security compliance.

4.2.1 Introduction to Languages, IDEs, Tools, and Technologies

The development environment provides the necessary tools, technologies, and platforms required for building and testing the proposed healthcare interoperability framework. Modern web development technologies and healthcare standards are utilized to ensure efficient system development and integration.

Development Tools

- Visual Studio Code (VS Code) for coding and development.
- Git and GitHub for version control and project management.
- Postman for API testing and validation.

Technologies Used

- React.js for frontend development.
- Node.js and Express.js for backend development.
- PostgreSQL and MongoDB for database management.
- HL7 FHIR for healthcare interoperability.
- RESTful APIs for communication between system components.

Development Platform

- Windows/Linux operating system.
- Modern web browsers for testing and deployment.
- Internet connectivity for accessing healthcare APIs and resources.

4.3 Testing Approaches

Testing is an important phase that ensures the proposed system functions correctly and meets the specified requirements. It is performed to verify the accuracy of terminology mapping, healthcare record generation, and system integration. Different testing techniques are used to identify errors and improve system reliability. Testing also helps validate security, performance, and interoperability features of the framework. The successful completion of testing ensures that the system operates efficiently and provides accurate healthcare data exchange.

4.3.1 Unit Testing

Unit testing verifies that individual functions, classes, or modules operate correctly in isolation. For this project, unit tests focus on preprocessing functions, technical indicator calculations, FinBERT sentiment extraction, and basic model behaviors.

a. Test Cases (Unit Testing)

Table No.7 Test Cases (Unit Testing)

Test Case ID	Test Scenario	Input	Expected Result
TC01	User Login	Valid Username & Password	User successfully authenticated and logged in
TC02	Terminology Search	Valid NAMASTE Term	Relevant terminology details displayed
TC03	ICD-11 Mapping	Selected NAMASTE Code	Corresponding ICD-11 TM2 code retrieved
TC04	Dual Coding	NAMASTE and ICD-11 Codes	Dual-coded diagnosis record generated

TC05	FHIR Bundle Generation	Patient & Diagnosis Data	Valid FHIR-compliant record generated
TC06	EMR Integration	Generated FHIR Record	Record successfully transmitted to EMR
TC07	API Validation	API Request	Correct response returned from server
TC08	Unauthorized Access	Invalid Credentials	Access denied and error message displayed

4.3.2 Integration Testing

The test results indicate that the proposed system successfully performed all major functionalities. Authentication, terminology search, code mapping, FHIR record generation, and EMR integration modules worked as expected during testing. The system generated accurate outputs and maintained secure communication between different components. No critical errors were observed during the testing process. Therefore, the overall performance of the system was found to be satisfactory and suitable for healthcare interoperability requirements.

Chapter 5

Results & Discussion

5.1 User Interface Representation

The User Interface (UI) of the proposed system is designed to provide a simple, interactive, and user-friendly environment for healthcare practitioners. It enables users to access various functionalities such as authentication, terminology search, code mapping, FHIR record generation, and healthcare record management. The interface focuses on ease of navigation and efficient interaction with the system.

Major User Interface Screens

- Login Page
- Dashboard
- NAMASTE Terminology Search Page
- ICD-11 TM2 Mapping Page
- FHIR Record Generation Page
- Healthcare Record Management Page
- Audit Log Page

The developed user interface ensures efficient access to system functionalities while maintaining usability, security, and consistency throughout the healthcare documentation process.

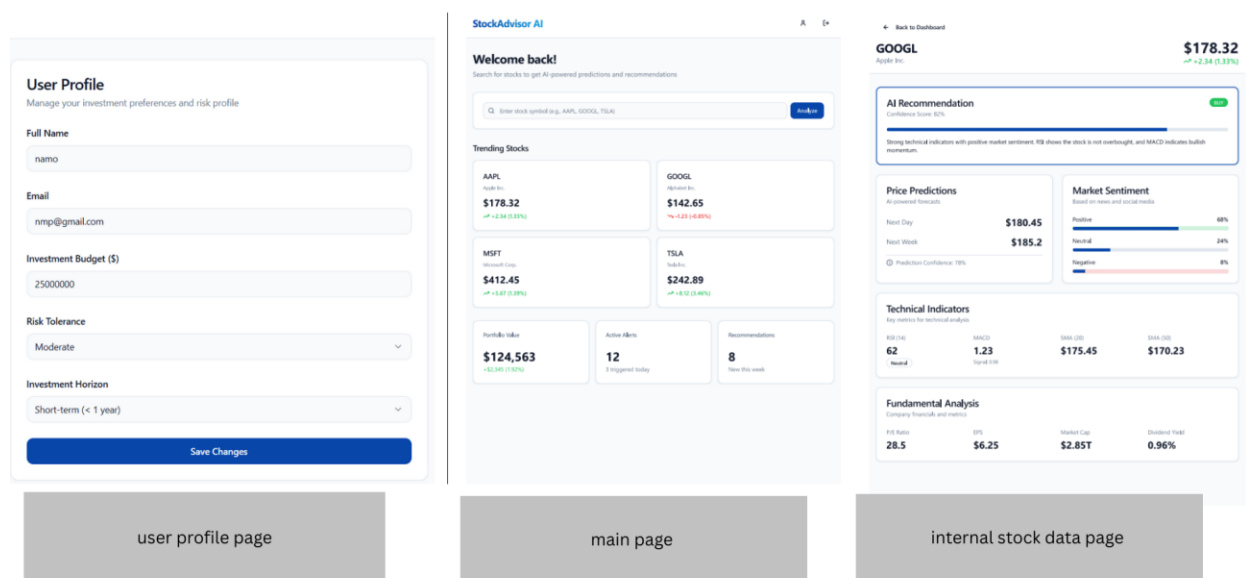


Fig 12 User Interface

5.1.1 Brief Description of Various Modules

The proposed healthcare interoperability framework consists of several modules that work together to provide seamless integration between NAMASTE and ICD-11 TM2 terminology systems. Each module performs a specific function to ensure efficient healthcare data management and interoperability.

Authentication Module

This module verifies user credentials and provides secure access to authorized users. It ensures data privacy and prevents unauthorized access to healthcare information.

Terminology Search Module

This module allows users to search NAMASTE terminologies and healthcare codes. It helps healthcare practitioners quickly access relevant diagnostic information.

Terminology Mapping Module

The mapping module establishes relationships between NAMASTE terms and corresponding ICD-11 TM2 concepts. It supports dual-coded healthcare documentation and improves interoperability.

FHIR Record Generation Module

This module generates HL7 FHIR-compliant healthcare records based on diagnosis and patient information. It ensures standardized healthcare data exchange.

EMR Integration Module

The EMR integration module enables secure transmission of healthcare records to Electronic Medical Record systems through APIs and interoperability standards.

Audit Logging Module

This module maintains records of system activities and transactions for monitoring, security, and compliance purposes.

The combination of these modules enables secure, standardized, and efficient healthcare data exchange between traditional AYUSH and modern biomedical healthcare systems.

5.2 Snapshot of System with Brief Description

This section presents the major interfaces of the proposed system along with a brief description of their functionality.

5.2.1 Login Page

The Login Page allows authorized users to access the system securely using valid credentials. It ensures authentication before granting access to healthcare services.

5.2.2 Dashboard

The Dashboard provides an overview of the system and allows users to navigate to different modules such as terminology search, mapping, and record management.

5.2.3 NAMASTE Search Page

This page enables users to search NAMASTE terms and retrieve relevant healthcare terminology information.

5.2.4 ICD-11 Mapping Page

The Mapping Page displays corresponding ICD-11 TM2 codes for selected NAMASTE terms and supports dual-coded diagnosis documentation.

5.2.5 FHIR Record Generation Page

This page generates HL7 FHIR-compliant healthcare records based on the selected diagnosis and patient information.

5.2.6 EMR Integration Page

The EMR Integration Page facilitates the transfer of healthcare records to Electronic Medical Record systems through secure APIs.

5.2.7 Audit Log Page

This page displays system activities and transaction records for monitoring, security, and compliance purposes.

5.2.8 User Profile Page

The User Profile Page allows users to view and manage their account information and system preferences.

5.3 Database Description

The database is an essential component of the proposed healthcare interoperability framework. It is used to store and manage healthcare terminologies, user information, mapping records, FHIR data, and audit logs. The database ensures efficient data retrieval, consistency, and secure storage of healthcare information.

User Database

Stores user credentials, profile information, and access permissions required for authentication and authorization.

NAMASTE Terminology Database

Contains standardized AYUSH terminology codes and related healthcare information used for terminology search and mapping.

ICD-11 TM2 Database

Stores ICD-11 TM2 concepts and classification details required for healthcare documentation and interoperability.

Mapping Database

Maintains relationships between NAMASTE terms and corresponding ICD-11 TM2 codes to support dual-coded diagnosis recording.

FHIR Records Database

Stores generated HL7 FHIR-compliant healthcare records and related patient information for integration with EMR systems.

Audit Log Database

Records system activities, user actions, and transactions for monitoring, security, and compliance purposes.

The database design ensures efficient healthcare data management and supports seamless communication between different components of the system.

5.3.1 Snapshot of Database Tables with Brief Description

This section presents the major database tables used in the proposed system along with their purpose and stored information.

Table 5.1: User Table

Field Name	Data Type	Description
User_ID	Integer	Unique identifier of user
Name	Varchar	Name of the user
Email	Varchar	User email address
Password	Varchar	Encrypted password
Role	Varchar	User role in the system

Description:

Stores user account information and authentication details required for secure system access.

Table 5.2: NAMASTE Terminology Table

Field Name	Data Type	Description
Term_ID	Integer	Unique term identifier
NAMASTE_Code	Varchar	NAMASTE terminology code

Term_Name	Varchar	Name of the healthcare term
Category	Varchar	AYUSH category

Description:

Stores standardized NAMASTE terminology data used for healthcare documentation and terminology search.

Table 5.3: ICD-11 TM2 Table

Field Name	Data Type	Description
ICD_ID	Integer	Unique ICD identifier
ICD_Code	Varchar	ICD-11 TM2 code
Disease_Name	Varchar	Disease description
Category	Varchar	Classification category

Description:

Contains ICD-11 TM2 concepts and classification information used for terminology mapping.

Table 5.4: Mapping Table

Field Name	Data Type	Description
Mapping_ID	Integer	Unique mapping identifier
NAMASTE_Code	Varchar	NAMASTE code
ICD_Code	Varchar	ICD-11 TM2 code
Mapping_Status	Varchar	Mapping status

Description:

Maintains relationships between NAMASTE terminologies and corresponding ICD-11 TM2 concepts.

Table 5.5: Audit Log Table

Field Name	Data Type	Description
Log_ID	Integer	Unique log identifier
User_ID	Integer	User performing action
Activity	Varchar	Activity performed
Timestamp	DateTime	Date and time of activity

Description:

Stores user activities and system transactions for monitoring, security, and compliance purposes.

5.4 Final Findings

The development and testing of the proposed healthcare interoperability framework demonstrated that effective integration between NAMASTE and ICD-11 TM2 terminology systems is achievable through standardized APIs and HL7 FHIR-based data exchange. The system successfully supported terminology search, code mapping, dual-coded diagnosis recording, and FHIR record generation.

The implemented framework improved healthcare documentation by enabling standardized representation of both traditional AYUSH and biomedical diagnoses within a unified environment. Secure authentication, database management, and interoperability features functioned as expected during testing.

The findings also indicate that the proposed system can facilitate seamless integration with Electronic Medical Record (EMR) systems, improve healthcare data consistency, and reduce manual efforts involved in terminology management. Overall, the framework provides a reliable, secure, and scalable solution for healthcare interoperability and supports the ongoing digital transformation of the healthcare sector.

Chapter 6

Conclusion & Future Scope

6.1 Conclusion

The proposed healthcare interoperability framework successfully addresses the challenge of integrating NAMASTE and ICD-11 TM2 terminology systems within a unified healthcare environment. By utilizing HL7 FHIR standards, RESTful APIs, and secure authentication mechanisms, the system enables standardized healthcare documentation and efficient healthcare data exchange.

The framework supports terminology search, code mapping, dual-coded diagnosis recording, and seamless integration with Electronic Medical Record (EMR) systems. The implementation and testing results demonstrate that the system operates effectively and provides accurate, secure, and interoperable healthcare services.

Overall, the proposed solution contributes to improving healthcare documentation, interoperability, and data consistency across traditional AYUSH and modern biomedical healthcare systems. It also supports digital healthcare initiatives by providing a scalable and standardized approach for healthcare information management.

6.2 Future Scope

The proposed healthcare interoperability framework provides a strong foundation for integrating traditional AYUSH and modern biomedical healthcare systems. However, several enhancements can be incorporated in the future to improve its capabilities and expand its applications.

Future Enhancements

- Integration with additional AYUSH systems and healthcare terminology standards.
- Implementation of Artificial Intelligence (AI) and Machine Learning (ML) for intelligent terminology mapping and clinical decision support.
- Development of multilingual support to improve accessibility for healthcare professionals across different regions.

- Integration with cloud-based healthcare platforms for improved scalability and accessibility.
- Addition of advanced healthcare analytics and reporting features for research and policy-making purposes.
- Support for mobile applications to enable healthcare services on portable devices.
- Enhancement of security mechanisms through advanced encryption and monitoring techniques.

Future Impact

These improvements can further strengthen healthcare interoperability, enhance digital healthcare services, and support the development of a more connected, efficient, and standardized healthcare ecosystem.

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